Unit 30 - Digital Graphics
Assignment 2 - Graphic Properties and Copyright

Introduction

Scenario / Vocational Context: *As part of your continued work for SuperSkills Studios the company would like you to demonstrate your understanding of the theory behind digital graphics, their properties, different file types and the legal implications of using and editing digital graphics. You are to do this by creating a document that covers the technical areas of working with digital graphics that demonstrates, to the best of your ability and understanding, your knowledge in this area.*

Task 1

(U30:P2) - Explain how different types of graphic images relate to file formats

Task description: *You are to write a small overview of digital graphics, covering file types, image conversion, good practice and the difference between vector and raster graphics.*

Different types of image file types covering jpeg and gif and their main uses

When creating a digital graphic, the first thing to keep in mind is what type of image file that the project should be saved as. Commonly, there are 5 main formats in which an image can be stored. However, there are many different image file types that can fall in this format. The 5 different formats consists of: TIFF, JPEG, GIF, PNG and Raw.

With TIFF (also known as TIF), which stands for Tagged Image File Format, images can be created into very large file sizes. This is due to the fact that TIFF images are uncompressed and contain a lot of detailed image data.

TIFFs are also extremely flexible in terms of colour due to their ability to use grayscale, CMYK for printing and RGB for image content and website use on a computer or device.

In graphics editing software, such as Photoshop, TIFF is the most common file type that is used. TIFF is also common in other page layout software such as Quark and InDesign, due to TIFF containing a lot of image data. TIFF files will be found to have “.tif” at the end of the file that it is used on.

With JPEG (also knows as JPG), which stands for Joint Photographic Experts Group, is the standard image type that is formatted. JPEG files are compressed images files so that it can store a lot of information in a small file, balancing the quality and quantity of the data.

However, JPEG image files are compressed in a way that will permanently remove the image detail (which is called lossy compression) when it is being compressed, in order to make the file such a small size.

Most digital cameras store photos in JPEG format, the reason being is so that the SD card could contain a lot more images when compared to uncompressed image files. Other than most digital cameras, JPEG files are usually used for images on the web. The reason for this is so that the small file size can easily be loaded on the webpage but without reducing the quality of the image.

On the other hand, a JPEG file wouldn't be good for line drawings, logos or graphics. This is also due to the lossy compression that this image file type has, because it makes them look noticeably pixelated which removes the quality of these graphics. JPEG files will be found to have ".jpg" at the end of the file that it is used on.

With GIF, which stands for Graphic Interchange Format, is a format that compressed the image just like the JPEG format but, different to the JPEG, the compression of the GIF is lossless. This means that no detail is lost in the compression, but the file can't be made as small as a JPEG would.

However, GIFs have an extremely limited colour depth, which is suitable for web use but not printing. This file format is never used for photography, this is due to the limited number of colours. GIF file types also have the ability to have a transparent image, so it won't have a white background like a JPEG would.

GIFs are most commonly used as an animation due to their ability to make a stop motion type file, creating an illusion that the image is being animated. GIF files will be found to have ".gif" at the end of the file that it is used on.

With PNG, which stands for Portable Network Graphics, is an image file that was created to replace GIF because the patient for GIF was owned by a single company and that nobody wanted to pay the licensing fees to use them. PNG also allows for a full range of colour depth and much better compression techniques.

PNG is used almost exclusively for web pages and never for printing pages. For photographs, PNG is not as good as JPEG because it creates a much larger file size. However, for images with some text, or line art, PNG would be better because it will appear much less pixelated than JPEG would.

For example, if a user screenshots on a computer or most commonly a Mac, the resulting image would probably be a PNG. This is due to the fact that most screenshots will be a mix of different images and text.

Raw image files usually contains data that if from a digital camera. These files are called raw because they haven't been processed and therefore can't be edited or printed yet. THere are a lot of different types of raw formats, because each different camera company often has it's own unique way of proprietary format.

Raw files usually contain a vast amount of data that is uncompressed. This is the reason that many raw files are extremely large. Usually, raw images are converted to TIFF before they are edited or colour-corrected.

How converting graphical images to different file types can alter an images file size, with examples showing the conversion from a RAW image to jpg and gif

The size of a RAW, JPEG and GIF image file is positively correlated with the resolution and images size, which is the total number of pixels, and the color depth, which is the bits per pixel. This also means that each of these file types are all raster images instead of vector, the difference of these two will later be explained.



This is an example of a RAW image that has been converted into a JPEG image.



This is an example of a RAW image that has been converted into a GIF image.

These images can be compressed in different ways, but it actually depends on what type of file the image is.In basic terms, this is mostly the reason that each different file type has a different file size, but there is a lot more to this.

For instance, if there is an image that has been compressed which stores either an exact representation or an approximation of the original image in a smaller number of bytes that can be expanded back to its uncompressed form with a decompression, this means that the file size might be the same as RAW.

This type of compression is called lossless, but this will also be later explained with more detail. GIF has a lossless type of compression, which means it can be uncompressed to an original size.

JPEG doesn’t have a lossless type of compression, so it will be impossible to get the original size back. This means that JPEG will already have a smaller file size after being compressed even when put into “original” size.

With some compression formats, images that are less complicated may result in smaller compressed file sizes. This sometimes results in a smaller file size for some lossless formats than lossy formats.

For example, graphically simple images such as images with large continuous regions like line art or animation sequences may be losslessly compressed into a GIF format and result in a smaller file size than a lossy JPEG format.

However, it is also likely that images with the same number of pixels and color depth can have very different compressed file size, this is due to the fact that each file type has a different method of compression.

For example, if there are two images that have exactly the same compression, number of pixels, and color depth for two images, different graphical complexity of the original images may also result in very different file sizes after compression due to the nature of the compression techniques.

At this point, the JPG should have a smaller file size. However, GIF is in normal use limited to an 8-bit palette or 256 colors, this means that there will be a smaller amount of bits per pixel.

GIF is most suitable for storing graphics with few colors, such as simple diagrams, shapes, logos, and cartoon style images, as it uses lossless compression, which is more effective when large areas have a single color, and less effective for photographic images.

Due to GIF's simplicity and age and its ability for animation capabilities, it can be so small of a file that it can have the same file size of the RAW if it had an animated version, but it also

depends on the quality. GIF also have a very low compression ratio compared to modern video formats.

Overall, converting a RAW image file to a GIF or JPG will definitely change the file size to a much smaller file size. The reason for this is due to the fact that GIF and JPEG already compress the image when they are converted, so it is automatically smaller than the full original raw image.

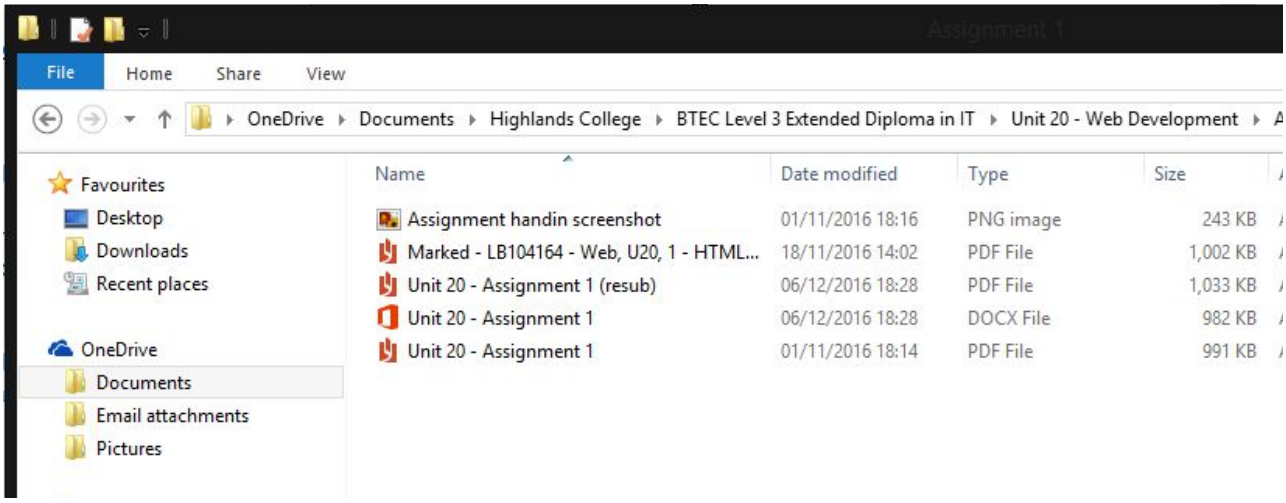
The JPEG file type's automatic lossy compression is the biggest reason why the file size of a JPEG image is smaller than a RAW image. Even though it's the biggest reason why, the lossy compression technique takes data from the image, meaning that it can be a lot smaller than the GIFs lossless compression.

The GIF file type's automatic lossless compression and its much lower colour pallette compared to RAW and JPEG make the file size a lot smaller than the RAW and JPEG, especially if JPEG had the same level of compression. Even though GIF may be smaller than a JPEG if it were compressed the same amount due to the GIF colour depth, JPEG will have a much more potential to decrease the file size compared to gif due to the lossy compression JPEG has and the lossless data the GIF has.

Good practice in image file management covering naming, folder structures and compression techniques (jpeg – lossy and gif – lossless)

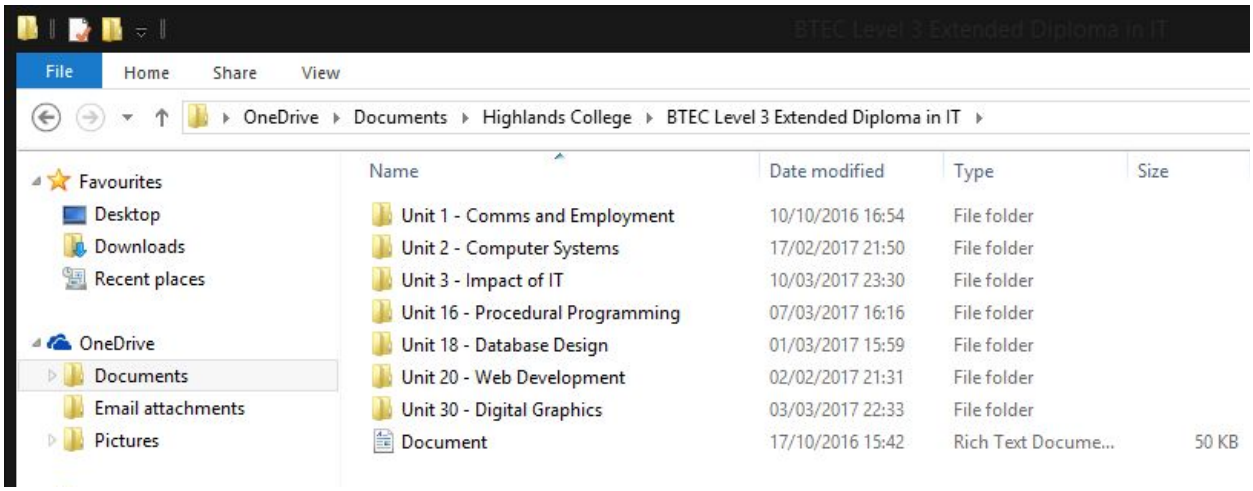
File management is a very important aspect when dealing with a computer. This is due to the fact that a good file structure will make it a lot easier to access work without the risk of losing it.

The first technique of file management is naming the files. It is important to name the files to relevant names so that it is self-explanatory for the user what the graphic consists of (i.e. mountain.jpg, so the viewer will know it is a JPEG image format of a mountain). Below is an example of good file names.



As you can see, the files consists of the assignment brief and the different document for the unit and the assignment number.

The second technique is to create a good folder structure. It is important that a folder structure is created in place so that the user can easily find the location of the graphical work and projects without time wasting. Below is an example of a good folder structure.



The third technique is to move the files. The reason for this is that being able to move the files is good for maintaining good organisation. For example, an archive file can be created which can contain the old work to be stored, rather than having to be left mixed in the same place with the new work. If it were all in one place, it would become congested with files, making the project that is being worked on at the time hard to find.

The fourth technique is to delete the unneeded files. The deletion of files maintains the organisation aspect as well as it allows the user to remove files to free up storage space as well as keep the folder organised for easy file tracking.

The fifth technique of file management would be using compression techniques, there are simple 2 types of file compression: Lossy and Lossless. Many different graphical image types has different compression techniques. For example, lossy compression file types will most commonly consist of JPEG and a lossless compression file type will consist of RAW, BMP and PNG.

When graphics are compressed through lossy compression, the file size will reduce as well as the quality of the graphic. This technique of compression is used in bitmap graphics as the reduction of the file size is done through losing the amount of pixels a graphic will have as the equation for file size is depending on the amount of pixels on the screen.

The quality of the graphic is lost because, obviously, the more the pixels that are present on a single surface are, the sharper the graphic will be. When the amount of pixels is lost, the space will need to be covered. This will cause the pixels to increase in size which will make them more visible, causing the image to be pixilated, reducing the quality of the image.

When a graphic is compressed through lossless compression, the file size will be reduced. However, this has an opposite effect when compared to lossy compression due to the quality of the image not being lost. This is done to the fact of the compression technique not actually removing data, instead it mathematically converts it for smaller file size.

For example, the image will consist of “RR,YYYYY,BBB”. With lossless compression, it will convert the image to “2R,5Y,3B”. This means that the data to represent the pixels inst lost like lossy compression, it is reduced in data value in order to reduce the file size of the image.

<u>The difference between vector and raster graphics</u>

In graphic design, there are 2 different ways in which an image can be created. Those methods are vector and raster/bitmap. Software such as photoshop creates and manipulates raster graphics, and Illustrator uses vector graphics.

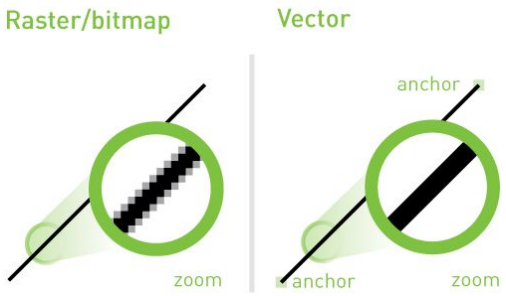
Raster graphics, or bitmap graphics, is a digital image which can be displayed on a screen which is made from pixels. Pixels are basically small points of colour that makes an image when put in mass numbers.

This method is ideal for creating an image on a screen as each pixel has its own colour value. Raster graphics are resolution dependent, which means that it can't be scaled larger without losing the quality of the image, making it appear pixelated.

Vectors can be used to create shapes, lines, polygons and any other shape imaginable, but as long as it has a line between two points. With this graphic, the software creates of it as a mathematical formula. For example, the user creates a line on a 45 degree angle from point A to point B for a distance of 50mm and is 2mm wide.

The software then draws the line based on this data rather than each pixel. Therefore vector graphics are infinitely scalable. If it is 50mm in length or 5 meters in length, the relationship between the points will always be the same.

This is why that vector graphics are always used for creating logos. For example, a designer can create the vector logo and that logo can then be applied to anything from a business card to building signage with no loss of quality. However, vector graphics can't be used to efficiently draw photo images.



In the image above it shows the difference between a line drawn using raster and vector. The vector graphic is infinitely scalable but the raster image become pixelated when enlarged.

Whilst these two methods are very different to each other, both Photoshop and Illustrator support them both. For example, vectors created in Illustrator can be imported into Photoshop, and Photoshop images can be placed in Illustrator.

Task 2

(U30:P6) - Explain the potential legal implications of using and editing graphical images

You are to explain what copyright law is, why it exists, what it protects, when something is deemed to have copyright.

Copyright is a type of legal protection which is given to content creators, giving them the ability of having specific rights that qualify for the protection of their content.

What I mean by this is that the owner of the copyright for a project can decide where the work is used and how it is distributed. Copyright generally lasts for the duration of the creator's life which can be +70 years, however it can be significantly lower.

The type of copyright law that would fall under graphical images is the I.P. rights, or intellectual property rights. Intellectual property rights is a type of law that protects ownership of creative work, such as artistic works, inventions, designs, etc. This law is very similar to the lay of

personal property and real estate, the reason is because it also protects the control of intangible assets.

The reason copyright exists is so that the creators of the creative piece of work can have a claim to their work just like their own house or other forms of property. If there wasn't a form of copyright, there will be so many types of plagiarism of the creator's work.

For example, if copyright didn't exist, there would be many different brands with the name "Microsoft" and "Apple", or other creators could copy books word for word and put their own name for it.

This would be catastrophic because the buyer will be confused on what will be the original product and won't be able to distinguish which one has the better quality or how to pinpoint their favourite product or creation to use for themselves.

The copyright laws will be varied on each continent and even each country, so it would be difficult to pinpoint what copyright protects in general. Nevertheless, the EU copyright laws can be used due to the fact that the EU copyright laws applies to every european country that is part of the EU, which the UK has very similar copyright laws to the EU.

In the EU and UK, copyright is a form of intellectual property law. This means that it protects original works of authorship including literary, dramatic, musical and artistic works. These artistic works can vary from poetry, novels, movies, song, computer software, website and architecture.

However, there are also certain things that the copyright cannot be applied to. These things can consist of facts, ideas, systems or methods of operation. Although, the copyright laws may protect the way these different things are expressed.

Obviously, to have something copyrighted, the creator will need to indicate that the piece of work is copyrighted in order for it to be valid. Since there are many ways in which a piece of work can be copyrighted, the example that will be used below will be how to indicate if an image is copyrighted.

Although there may not always be an immediate way to detect if an image has been copyrighted, a few telltale sign do exist. Some of the most simple ways of filtering out protected images include checking for watermarks, copyright symbol, a note indicating ownership or sourcing, official copyright listing in the Copyright office and that it just simply isn't your creation.

With watermarks, which are overlays of a symbol or text, may be hidden covertly or displayed upfront to protect an image from being used without owner's approval. Removing a watermark from someone else's property is possible, yet highly discouraged.

Small details such as a copyright symbol located on the image and a note indicating ownership or sourcing of the image and an official copyright listing in the Copyright office are less obvious than a watermark.

However, it is important to find small copyright symbols and small notes to check if the image that is found is copyrighted. Otherwise the legal implications of copyright will be applied because an offence has been committed if the image was used.

If none of the above is found, it still will be a dangerous act on trying to use the image, so it may seem that the images aren't "officially" copyrighted. The reason being is that at the creating of an image or the snap of a picture the work has immediately become copyrighted and the owner is the only person with a legal right to distribute, replace or display the work.

Overall, for safe measures, it would be best not to utilize a creator's image unless the creator states that the image can be used by others or that a message should be sent to the creator whether it is allowed to be used or not.

<u>You are also to explain how one may find copyright free images and what these are called as well as other methods of avoiding copyright infringement.</u>

Even though there are copyright laws for the work that has been created from the creator himself, there sometimes are pieces of work that are copyright free. These copyright free pieces of work are usually called "Royalty-free".

Royalty-free can be defined as a piece of work that no longer has a copyright if the copyright has expired or because the copyright holder has specifically given up or transferred some or all right to the property in question.

What I mean by this is that copyright free is when the copyright duration of a piece of work has been expired, the owner has given up the ownership of the copyright or just never claimed copyright for the work in the first place.

There are also ways in which copyright infringement can be avoided, such as gaining ownership of copyright. However, gaining ownership of copyright varies depending on circumstances. In the UK, the creator/author is automatically the first owner of the copyright of the piece of work. Copyright is considered a form of property, like a physical property, so it can be transferred.

For example, if someone is working for a company, the copyright for any material that has been created is automatically given to the company, as this is generally part of the contract.

There are also other ways in which copyright can be avoided. These methods can consist of: parody, fair dealing/fair use doctrine, critique,review and current affairs.

With parody, the use of another person's work is permitted if it is used for creating a parody. A parody is basically a form of imitation of another person's piece of work, in order to make a joke out of it or mock it.

With fair dealing/fair use doctrine, in the UK, the viewer may use a copyrighted work if it is considered "Fair dealing". This is determined by the court and is based on the economic impact that the user would have on the copyright holder of the work. If the economic impact on the copyright holder is deemed insignificant, then the usage may count as fair dealing. The intellectual property office UK states: "Where the economic impact is not significant, the use may count as fair dealing".

With Critique/Review, if a piece of work is being critiques or being reviewed, such as an advert or another graphical piece of work, then it would be legal to showcase the work in the review. However, using the whole piece of work isn't allowed for some pieces of work, such as a video or movie.

With current affairs, if the work displays something that is a current affair, such as a picture of a specific natural disaster or the work itself is a current affair. By then it would be allowed to use it as part of a report, such as a news report, in order to help report the affair.

Lastly you are to explain the possible legal implications of failing to adhere to copyright law.

If someone's copyrighted work is infringed on, the offender could receive a polite letter or email from the author asking the offender to remove the material from their ownership or otherwise stop the use of the work.

The offender could get a notice demanding to cease and desist using and distributing the copyrighted work, or the offender might discover that the Internet service provider, if on a website, has already removed infringing content or that the police have confiscated the copyrighted work from the offender

It is possible that the offender could face penalties of more than £100,000 for each infringed work. By then it is very possible that the offender could go to jail. In general, It all depends on the extent of the infringement.

There are generally 4 different levels of copyright policing law, those 4 levels consist of: Isolated infringement, digital millennium copyright act notice, infringement suit and criminal case.

With Isolated infringement, if a user found a cool photo that he republished on the his website or an excellent blog the user reposted, the user is most likely violated somebody's copyright.

This could be “innocent infringement” if the user didn’t realize the work was copyrighted. Innocent infringement, however, is becoming harder to claim as writers and photographers are becoming more savvy about placing copyright notices on their work.

Even so, if the copyright holder discovered the infringement, the user will likely receive only a letter or email asking him to remove the work. If the user complies, the matter’s closed.

With a digital millennium copyright act notice, when a copyright owner discovers a serious infringement on the Internet, and it’s up to the owner to decide what’s serious, she can send a Digital Millennium Copyright Act notification to the Internet service provider hosting the site demanding that the infringing material be removed.

A DMCA takedown notice specifies where on a site the infringing content is placed so the ISP can take down just that material.

However, if the offender has multiple occurrences of copyright infringement, the ISP could take down the whole site to protect its own “safe harbor” status, which prevents it from being charged with hosting infringing works.

The offender can dispute the claim in the DMCA notice on a basis of fair use, but if the offender posted a complete article, image or song on the offender’s site without permission, his argument will likely not hold up.

With an infringement suit, the owner of a work’s copyright can sue the offender for copyright infringement, seeking damages from lost sales because of the offender’s use or distribution of the material.

If the work is registered with a Copyright Office from the country that the creator lives in, the copyright owner can also seek legal costs and statutory damages of up to £150,000 for each work a court finds was willfully infringed upon. If the offender profited from illegal sales of the copyright-protected work, the court can award that income to the copyright owner.

With a criminal case, if an offender’s alleged infringement is serious enough the offender could face criminal charges. However, only if it were in the most serious of cases when dealing with copyright, whether it is from mass profit from the copyrighted work or that there are many pieces of copyrighted work.

For example, if an offender maintained a website for unauthorized copying and distribution of multiple copyrighted works for “commercial advantage or private gain”, it is most likely that the offender will face criminal charges from the creators and even the police.

In much more serious cases, if an offender made any income off the pirated work, the offender would likely forfeit that as well. Anyone convicted of criminal copyright infringement faces a maximum of 10 years in prison, a £250,000 fine or both.

Task 3

(U30:D2) - Discuss the impact that file format, compression techniques, image resolution and colour depth have on file size and image quality

Task description: *You are now to discuss the different aspects and technicalities of digital graphics and how these affect file size and image quality. You must use concise terminology to convey your understanding in this area.*

Introduction

In this documentation I will be explaining why user must know how to use and notice the different file compressions, different file formats, different image resolutions and also the different colour depths.

For each category that will be mentioned I will be showing and explaining the different aspects in file sizes that each of these topics have and compare them with one another. The file formats that will be presented in the documentation will consist of: PSD, GIF, and JPEG.

How different file types can impact file size and quality by looking at jpg and gif (ensuring you cover the technical details of each file type).

With JPEG (Joint Photographic Experts Group), this file format is an image file format and can be used to store digital bitmap images. A bitmap image consists of tiny little squares which are called pixels, which are put together like a grid forming an image.

This file formats consists of characteristics such as compatibility, file size and even upload and download high speeds, which practically make it both internet friendly and convenient.

This file format very advantageous for web use due to its ability for lossy compression while still keeping a good quality that is still considered pleasing. JPEG also has characteristics that give it great image qualities such as “CMYK” which basically gives a JPEG file format better quality for printing out paper.

This is an advantageous quality when compared to many other formats because not all formats have this ability. Another advantage to JPG is that this file format is so widely used it can be opened on almost any software including Photoshop and Fireworks.

In the other hand, this file format does have a noticeable disadvantage, in which it can take a lot more storage space if the size of the image resolution is increased, which can occupy a lot of storage space and increase the upload time to an external storage or the web. However, this can be overcome with compression techniques

A very significant quality disadvantage of JPEG is that when the image is enlarged, the pixels become more visible making the quality of the picture decrease. However, if the picture is made smaller then a vice versa effect takes place making the image have an aspect of better quality.

With GIF (Graphic interchange format), this file format is classed as a bitmap format but is most suited to the internet and is also most commonly found on internet images. The properties that this file format has are compatibility and don't take too long to load up on the internet and also doesn't take a long time to load up on the webpage sites.

This file format is best used with solid colours and also animations on websites, but the disadvantages to this file format that it only uses the colour schemes grayscale and "RGB" which limit its image quality.

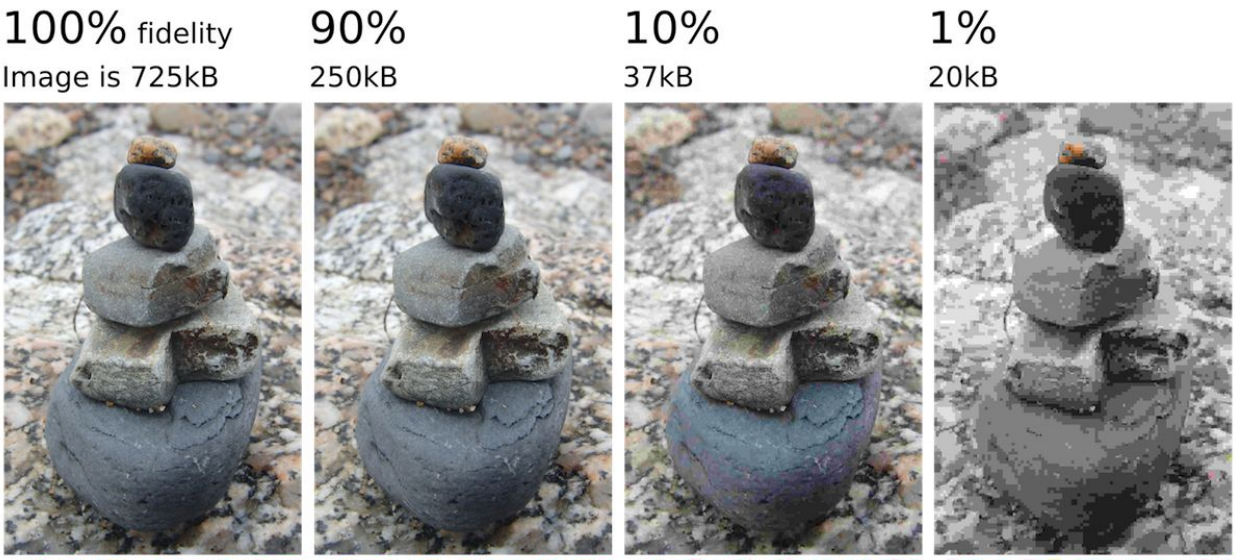
However, another problem with this format is that it would be, due to the vast development and sophistication of the programme Photoshop, it wouldn't have the potential of being as high quality as JPEG due to the very limited colour depth of the GIF file format.

<u>How different compression techniques can impact file size and quality by using one example for both lossy and lossless compression.</u>

As mentioned previously, there are two methods when it comes to file compression and that they are lossy and lossless. Below I will be mentioning what these two methods do to impact the file size and the quality of the image.

When using lossy compression, this method has a huge effect on the image. The reason for this is due to the fact that lossy compression removes the data of the image to make it a much smaller size, therefore decreasing the quality of the image itself.

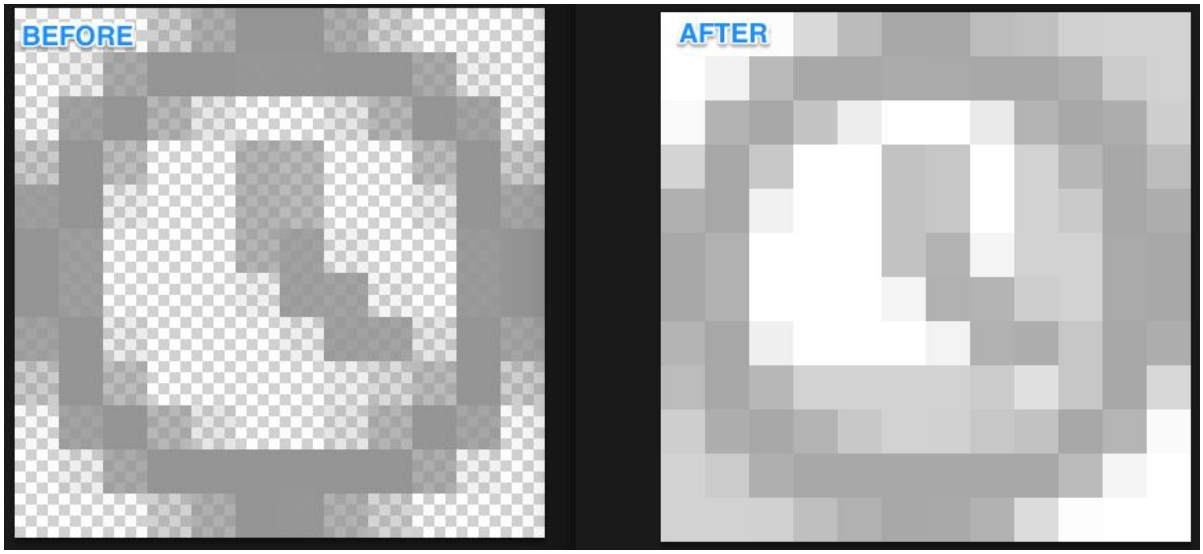
The reason that loss of data leads to the image in having such a bad quality is due to the pixelation that will be much more visible to the viewer. This method of compression may have its disadvantages, however it does compress a lot more significantly when compared to most forms of lossless compression.



This is an example of lossy compression of an image with the JPEG file format.

When using lossless compression, this method is better in quality than the method that I mentioned above as it does not remove the initial quality of the image. However, due to the compression technique not actually removing the data, this method of compression will still be a larger file size when compared to lossy compression.

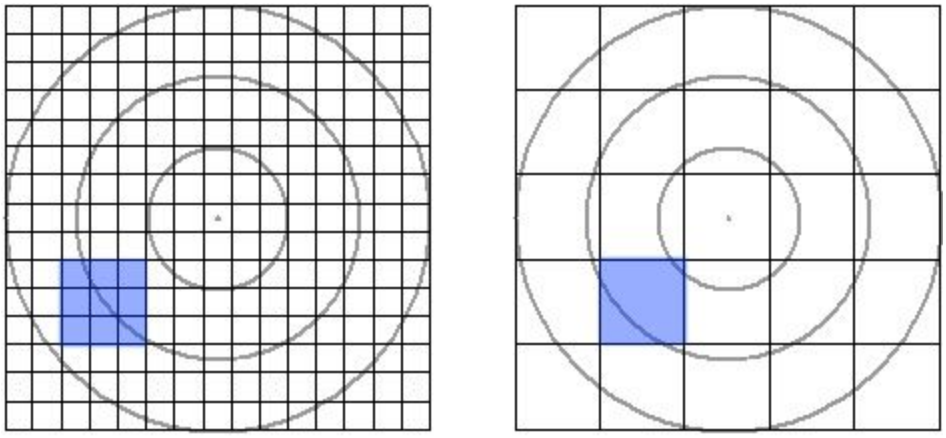
The reason for this is that, as I mentioned before, lossless compression uses a mathematical method to reduce the amount of data, like the example “RR,YYYYY,BBB” to “2R,5Y,3B”. While lossy compression just removes the data that is used to build the image, hence the reason that JPEG is much more common than PDF files.



This is an example of lossless compression of an image with the PNG file format.

What image resolution is and how this can affect file size and quality using visual examples.

The way that image resolution is measured is by the number of dots there are on a printed image on paper, the amount of dots on a printed piece of paper is measured in DPI (dots per inch), which means how much the quality of the image is. An example of this would be if an image printed with lots of dots and colour then the image will have a better quality when compared to an image that was printed a black and white image. Below is an example of the DPI a printer with high resolution compares to one with a low resolution.



High Resolution **Low Resolution**

Image resolution can also be used for measuring graphical displays, such as an image or video, via the number of pixels that make up the image. In relation to file formats it should be evidently clear that the larger the image the better resolution and image quality the image will have; vice versa for when the image is smaller, resolution is also decided via the size of the image in question.

An effect that a high resolution image might have is that the larger the image, then the more room it takes up on hard-drive storage which basically gives less storage space. Below is an example of different image resolutions varying from 240p to 1080p which can be applied for images and videos.

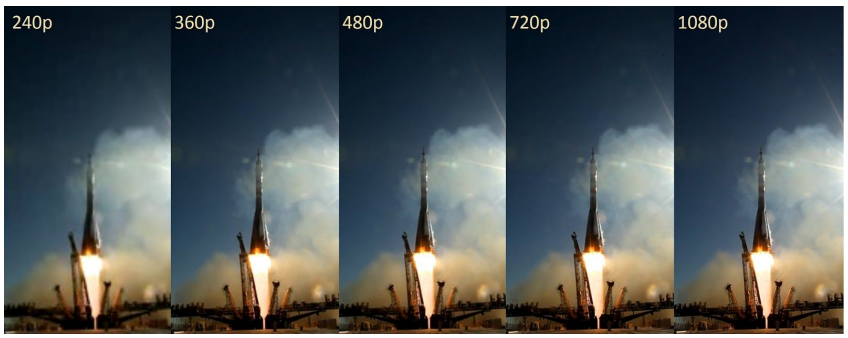


Image resolution can also be used to measure the amount of fixed pixels on a computer monitor display. Just like printing dots on paper, they cannot be physically moved.

An effect that this might have with graphical work is that the screen is limited to the amount of resolution that it's possible to see. For example, if there were 2 screens that is 1920x1080 (1080p) and another with 3840x2160 (4K UHD). Even though both screen can display any image at any resolution, one will be more noticeable than the other to the human eye.

For instance, the 1080p screen will only be able to display images that are 1080p and below with it being noticed by the user. This can also be applied to the 4K screen, but with a much more significant resolution image of 4K and below.

Overall, resolution is a fixed measurement. Which is why it is used on printing paper, videos and images that cannot be manipulated to a larger size and the screen's pixel count. The reason that resolution is used on images and videos is that it would be virtually impossible to increase the resolution count of an image and a video.

In order to get a better quality image or video, a new one will have to be found at a much higher resolution. Which is why video sharing websites such as youtube refresh the whole download of the video as it is playing when a new resolution is chosen.

<u>What colour depth is and how this can affect file size and quality.</u>

The more colour an image has, the more attractive and more detailed an image will be presented. Colour depth is the term that is used to describe the amount of colour used in an image. Furthermore, there are many different colour depths which consist of monochrome, 16 bit colour, 24 bit colour and 32 bit colour.

With 1-bit colour depth, which is also called monochrome, it is probably the easiest and simplest form of the colour depth group. Basically, this consist of only 2 shades: Black and white. It creates an image by mixing white and black in a way to produce an optical illusion to make it looks like there are different shades of gray, producing a visual image.

The reason that it only has black and white colours is because the pixels in this colour depth could either be turned on with white light or turned off. This means that this type of colour depth is a 1-bit colour depth, which consists of 2 different pieces of data: on and off.

With 16-bit Colour, which is also called High colour, which gives the ability for an image to process as many as 65 thousand (or 65,536) colours to a monitor. Even though this colour depth does contain a significant amount of colours, this type of colour depth would be in the medium range of image quality since the colours in 16 bit doesn't completely replicate the image, which can become noticeable when compared to higher colour depths.

The reason that this colour depth has higher quality and is coloured unlike the previous colour depths is due to the bytes that it utilises, one of the bits is used for the actual colour itself and the other bit is used for the shading of the colour in the image produced on screen.

With 24-bit Colour, which is also called True colour, this gives the ability for an image to process as many as 16.7 million (or 16,777,215) different colour palettes that is displayed on a monitor. This colour depth has a colossal amount of colour palettes, which can be noticeable to 16-bit colour depth in the colour quality. This colour depth has a better quality than the previous one is that this uses a lot more bytes for pixel in the display system, which is 8 bits for each red, green and blue subpixel.

The reason that this colour depth is called True colour is due to the fact that all the colours that are represented in a 24-bit image equally represents the colour to the original source to the human eye. It is said that the human eye can discriminate up to ten million colours, since 24-bit is 16.7 million colours, it means that it contains all the colours that a physically visible to a viewer.

With 32-bit Colour, which is also called True colour. However, there is a difference even though this format uses the same format above for the Red,Green and Blue colours but it also stores transparency information for each pixel.

This is so it allows each of the 256 pixels to be fully opaque to fully transparent which is very important when it comes to having a good quality image, however it does man the storage size for the image will be much larger.

Conclusion

In conclusion, as my document shows there are some file formats that are more suited to the web than others due to their characteristics which enable them to viewable on the web easier and also to be uploaded or downloaded much more efficiently with less storage space being used.

Compression techniques can also be very useful whilst viewing files on the web making them upload at a much more significant speed and providing convenience for the viewer of that designated image with less storage being used.

When it comes to colour depth it is very important to know to improve the image so it would be enhanced to be aesthetically pleasing to the viewer due to the more colour it has. This will then increase the file size which also ties into image quality, the reason being is that if the image is enlarged then the file will become larger in storage size but the picture quality which will also increase with that extra data.

In my personal opinion, I believe that it is very important to know all these different qualities in regards to image editing and viewing images on the internet due to their ability to improve the upload speeds, qualities, compatibility, and the graphics aesthetically pleasing aspects to the viewer.

Overall, the highest priority to understand all of these aspects is to first understand which file types are best suited for the purpose of the graphic. If this is not understood, then it would be a nightmare on trying to find out what is the best suited file type for the image that is being worked on.